

Effect of tart cherry juice (*Prunus cerasus*) on melatonin levels and enhanced sleep quality.



BACKGROUND: Tart Montmorency cherries have been reported to contain high levels of phytochemicals including melatonin, a molecule critical in regulating the sleep-wake cycle in humans. **PURPOSE:** The aim of our investigation was to ascertain whether ingestion of a tart cherry juice concentrate would increase the urinary melatonin levels in healthy adults and improve sleep quality. **METHODS:** In a randomised, double-blind, placebo-controlled, crossover design, 20 volunteers consumed either a placebo or tart cherry juice concentrate for 7 days. Measures of sleep quality recorded by actigraphy and subjective sleep questionnaires were completed. Sequential urine samples over 48 h were collected and urinary 6-sulfatoxymelatonin (major metabolite of melatonin) determined; cosinor analysis was used to determine melatonin circadian rhythm (mesor, acrophase and amplitude). In addition, total urinary melatonin content was determined over the sampled period. Trial differences were determined using a repeated measures ANOVA. **RESULTS:** Total melatonin content was significantly elevated ($P < 0.05$) in the cherry juice group, whilst no differences were shown between baseline and placebo trials. There were significant increases in time in bed, total sleep time and sleep efficiency total ($P < 0.05$) with cherry juice supplementation. Although there was no difference in timing of the melatonin circadian rhythm, there was a trend to a higher mesor and amplitude. **CONCLUSIONS:** These data suggest that consumption of a tart cherry juice concentrate provides an increase in exogenous melatonin that is beneficial in improving sleep duration and quality in healthy men and women and might be of benefit in managing disturbed sleep.